

Technical Report: Diagnostics of Acoustic Classification Errors and Comparative Literature Review

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1. Error Analysis

While top-tier models achieved near-perfect accuracy due to the dataset’s overlap strategy, examining the failure modes of the lower-capacity **Custom CNN** provides a highly valuable diagnostic window into the inherent acoustic difficulties of underwater vessel classification.

1.1. Primary Misclassification Patterns

1. KaiYuan Misclassified as CargoShip
Both <i>KaiYuan</i> (passenger/ferry) and <i>CargoShip</i> represent massive, heavy-tonnage vessels utilizing low-RPM diesel propulsion. These engines concentrate acoustic energy in the ultra-low frequency bands (10–500 Hz). Because their spectral envelopes and fundamental propeller blade rate frequencies overlap significantly, base CNNs struggle to identify discriminative features.
Proposed Engineering Fix
Implement higher-resolution, narrowband spectrograms specifically focused on the 0–1000 Hz range to extract cylinder-firing rates rather than broadband Mel-bins.
2. Uuv Misclassified as CargoShip
This error is a byproduct of the 97% class imbalance . <i>Uuvs</i> use quiet, high-frequency electric propulsion with faint acoustic footprints. When the model encounters a low-amplitude signal buried in noise, it defaults to predicting <i>CargoShip</i> to minimize global categorical cross-entropy loss.
Proposed Engineering Fix
Deploy severe dataloader undersampling for the majority class and utilize SpecAugment (time/frequency masking) on the minority classes to artificially boost representation.
3. SpeedBoat Misclassified as Noise
<i>Speedboats</i> generate intense high-frequency broadband cavitation noise. However, high-frequency waves attenuate rapidly in water. In far-field recordings, the signature fades into the ambient high-frequency soundscape (wind, surface waves), causing the model to fail in detection.
Proposed Engineering Fix
Apply a pre-whitening filter or adaptive background noise subtraction to enhance the Signal-to-Noise Ratio (SNR) and flatten the ambient noise floor.

2. Literature Comparison

The following table contextualizes the **ShipSense AI** project against five landmark papers in the field of under-water acoustic classification.

Table 1: Comparative Analysis with State-of-the-Art (SOTA) Literature

Paper	Year	Dataset	Method	Acc.	F1	Key Contribution / Notes
Santos-Domínguez	2016	ShipsEar	GMM/SVM	75.4%	0.74	Original baseline paper; proved MFCC viability in marine domains.
Irfan et al.	2021	DeepShip	CNN/AE	97.2%	0.96	Evaluated massive 47-class dataset; used robust CNN architectures.
Ke et al.	2018	ShipsEar	2D CNN	93.6%	0.92	Pioneered the translation of 1D audio to 2D image formats for deep learning.
Shen et al.	2020	ShipsEar	CRNN	94.8%	0.93	Demonstrated LSTMs' efficacy in capturing long-term temporal dependencies.
Ren et al.	2022	DeepShip	AST	98.1%	0.97	Utilized self-attention (Transformers) to capture global spectral context.
ShipSense AI	2026	ShipsEar	ResNet-18	99.9%*	0.99*	Utilized transfer learning; metrics inflated by 50% segment overlap.

**Note: Performance metrics for ShipSense AI reflect current dataset constraints regarding parent-file leakage.*